

DSC 140A

Probabilistic Modeling & Machine Learning

Lecture 11 | Part 1

Probabilistic Modeling

Probabilistic Modeling

- ▶ Where does data come from?
- ▶ We might imagine that “Nature” generates it using some random (i.e., **probabilistic**) process.
- ▶ Maybe modeling this probabilistic process will suggest new ways of making predictions?

Example: Flowers

- ▶ Suppose there are two species of flower.
- ▶ One species tends to have more petals.
- ▶ **Goal:** Given a new flower with $X = x$ petals predict its species, Y .



Example: Flowers

- ▶ **Idea:** The number of petals, X , and the species, Y , are **random variables**.
- ▶ **Assumption:** When Nature generates a new flower, it picks X and Y from some **probability distribution**.
- ▶ Let's imagine (for now) that we know this distribution.

The Joint Distribution

- The **joint distribution** $\mathbb{P}(X = x, Y = y)$ gives us full information.¹

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

¹Note: it's conventional to encode Y as 0 or 1 instead of 1 and -1.

Observation

- ▶ The entries of the joint distribution table sum to 100%.
- ▶ Mathematically: $\sum_{x \in \{0,1,\dots,6\}} \sum_{y \in \{0,1\}} \mathbb{P}(X = x, Y = y) = 1.$

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

Marginal Distributions

- What is the probability that a new flower has $X = 4$ petals (regardless of species)?

	$Y = 0$	$Y = 1$
$X = 0$	0%	0%
$X = 1$	5%	0%
$X = 2$	10%	5%
$X = 3$	15%	15%
$X = 4$	5%	20%
$X = 5$	0%	15%
$X = 6$	0%	10%

Marginal Distributions

- ▶ The **marginal distribution** for X is found by summing over values of Y .
- ▶ That is: $\mathbb{P}(X = x) = \sum_{y \in \{0,1\}} P(X = x, Y = y)$

	$Y = 0$	$Y = 1$
$X = 0$	0%	0%
$X = 1$	5%	0%
$X = 2$	10%	5%
$X = 3$	15%	15%
$X = 4$	5%	20%
$X = 5$	0%	15%
$X = 6$	0%	10%

joint

$X = 0$	0%
$X = 1$	5%
$X = 2$	15%
$X = 3$	30%
$X = 4$	25%
$X = 5$	15%
$X = 6$	10%

marginal in X

Marginal Distributions

- What is the probability that a new flower is species 1 (regardless of number of petals)?

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

Marginal Distributions

- ▶ The **marginal distribution** for Y is found by summing over values of X .
- ▶ That is: $\mathbb{P}(Y = y) = \sum_{x \in \{0, \dots, 6\}} P(X = x, Y = y)$

	$Y = 0$	$Y = 1$
$X = 0$	0%	0%
$X = 1$	5%	0%
$X = 2$	10%	5%
$X = 3$	15%	15%
$X = 4$	5%	20%
$X = 5$	0%	15%
$X = 6$	0%	10%

joint

$Y = 0$	35%
$Y = 1$	65%

marginal in Y

Observation

- ▶ The probabilities in the marginal distributions also sum to 1.

Exercise

Suppose flower A has 4 petals. What do you think its species is?

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

Intuition

- It seems **more likely** that a petal with 4 flowers is from species 1.

	$Y = 0$	$Y = 1$
$X = 0$	0%	0%
$X = 1$	5%	0%
$X = 2$	10%	5%
$X = 3$	15%	15%
$X = 4$	5%	20%
$X = 5$	0%	15%
$X = 6$	0%	10%

Conditional Probabilities

- This is captured by the **conditional probability**
 $\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(X = x, Y = y) / \mathbb{P}(X = x).$

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

joint

$\mathbb{P}(Y = y \mid X = 1)$	
Y = 0	100%
Y = 1	0%

$\mathbb{P}(Y = y \mid X = 2)$	
Y = 0	66.5%
Y = 1	33.3%

$\mathbb{P}(Y = y \mid X = 4)$	
Y = 0	20%
Y = 1	80%

Conditional Probabilities

- The **conditional probability**

$$\mathbb{P}(X = x \mid Y = y) = \mathbb{P}(X = x, Y = y) / \mathbb{P}(Y = y).$$

	Y = 0	Y = 1
X = 0	0%	0%
X = 1	5%	0%
X = 2	10%	5%
X = 3	15%	15%
X = 4	5%	20%
X = 5	0%	15%
X = 6	0%	10%

joint

$\mathbb{P}(X = x \mid Y = 0)$	
X = 0	0%
X = 1	14.2%
X = 2	28.5%
X = 3	42.8%
X = 4	14.2%
X = 5	0%
X = 6	0%

Observation

- ▶ Conditional probabilities sum to 1 as well.
- ▶ For any fixed x :

$$\sum_y \mathbb{P}(Y = y \mid X = x) = 1$$

- ▶ For any fixed y :

$$\sum_x \mathbb{P}(X = x \mid Y = y) = 1$$

Five Distributions

- ▶ We've seen five distributions:
 - ▶ **Joint**: $\mathbb{P}(X = x, Y = y)$
 - ▶ **Marginal in X**: $\mathbb{P}(X = x)$
 - ▶ **Marginal in Y**: $\mathbb{P}(Y = y)$
 - ▶ **Conditional on X**: $\mathbb{P}(Y = y \mid X = x)$
 - ▶ **Conditional on Y**: $\mathbb{P}(X = x \mid Y = y)$
- ▶ If we know the **joint** distribution, we can compute any of the others.

Bayes' Theorem

- **Bayes' Theorem** relates conditional probabilities and provides another way of computing them:

$$\mathbb{P}(Y = y \mid X = x) = \frac{\mathbb{P}(X = x \mid Y = y)\mathbb{P}(Y = y)}{\mathbb{P}(X = x)}$$

Bayes' Theorem

► Derivation:

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Lecture 11 | Part 2

Bayes Decision Theory

Probabilistic Classification

- ▶ We have seen several **paradigms** for classification:
 - ▶ Nearest neighbors
 - ▶ Linear predictors
- ▶ Now we see a **probabilistic** paradigm.

Probabilistic Classification

- ▶ In ML, we think of X as a feature and Y as a label.
 - ▶ **Example:** Given new flower with $X = x$ petals, predict the species, Y .
- ▶ **Idea:** predict the **most likely** label y given $X = x$.
 - ▶ I.e., predict the y that maximizes $\mathbb{P}(Y = y \mid X = x)$.

Binary Probabilistic Classification

- ▶ Predict 1 if $\mathbb{P}(Y = 1 \mid X = x) > \mathbb{P}(Y = 0 \mid X = x)$; otherwise predict 0.
- ▶ That is, pick whichever label is more likely given the features.

Bayes Classification Rule

- ▶ This is the **Bayes (binary) classification rule**:
 - ▶ Predict class 1 if $\mathbb{P}(Y = 1 \mid X = x) > \mathbb{P}(Y = 0 \mid X = x)$;
 - ▶ Otherwise, predict class 0.

Bayes Decision Theory

- ▶ Two equivalent forms of the Bayes classification rule:
- ▶ **“Original” form:**
 - ▶ Predict class 1 if $\mathbb{P}(Y = 1 \mid X = x) > \mathbb{P}(Y = 0 \mid X = x)$;
 - ▶ Otherwise, predict class 0.
- ▶ **“Alternative” form:**
 - ▶ Using Bayes’ rule,
$$\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(X = x \mid Y = y)\mathbb{P}(Y = y) / \mathbb{P}(X = x) \dots$$
 - ▶ Predict class 1 if
$$\mathbb{P}(X = x \mid Y = 1)\mathbb{P}(Y = 1) > \mathbb{P}(X = x \mid Y = 0)\mathbb{P}(Y = 0)$$
 - ▶ Otherwise, predict class 0.

Main Idea

If we know the conditional probability of the label Y given feature X , the Bayes classification rule is a natural way to make predictions.

Optimality

- ▶ We'll see that the Bayes classification rule is **optimal** in a certain sense.

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Lecture 11 | Part 3

Continuous Distributions

Example: Penguins

- ▶ Suppose there are two species of penguin.
- ▶ One species tends to have longer flippers.
- ▶ **Goal:** given a new penguin with flipper length $X = x$, predict its species, Y .

Five Distributions

- ▶ In this situation, what do the five distributions look like?
 - ▶ Joint distribution of X and Y
 - ▶ Marginal distribution in X
 - ▶ Marginal distribution in Y
 - ▶ Conditional on X
 - ▶ Conditional on Y

Marginal in Y

- ▶ What is the probability that Nature generates a penguin from species Y ?
 - ▶ Marginal distribution: $\mathbb{P}(Y = y)$.
- ▶ This is a **discrete** distribution, as before.
- ▶ Example:

$Y = 0$	30%
$Y = 1$	70%

Marginal in X

- ▶ What is the probability that Nature generates a flipper length of x , without regard to species?
- ▶ Flipper length is a **continuous** random variable.
- ▶ Distribution is described by a **probability density function (pdf)**, $p : \mathbb{R} \rightarrow \mathbb{R}^+$.

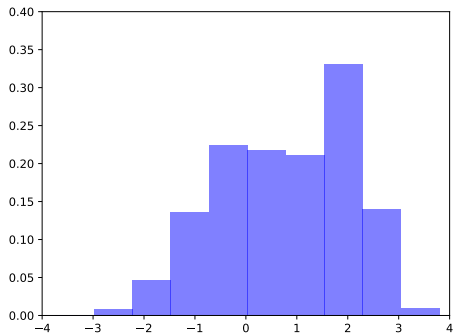
Recall: Density Functions

- ▶ A **probability density function (pdf)** for a random variable X is a function $p : \mathbb{R} \rightarrow \mathbb{R}^+$ satisfying:

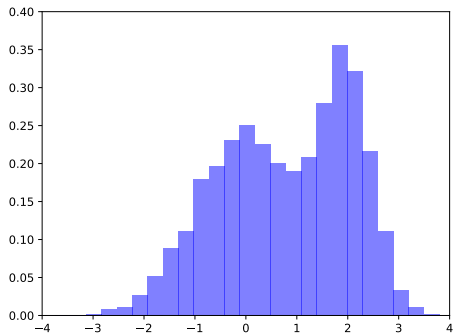
$$\mathbb{P}(a < X < b) = \int_a^b p_X(x) dx$$

- ▶ That is, the pdf p describes how likely it is to get a value of X in any interval $[a, b]$.
- ▶ Note: $\int_{-\infty}^{\infty} p_X(x) dx = 1$, but $p(x)$ can be larger than one.

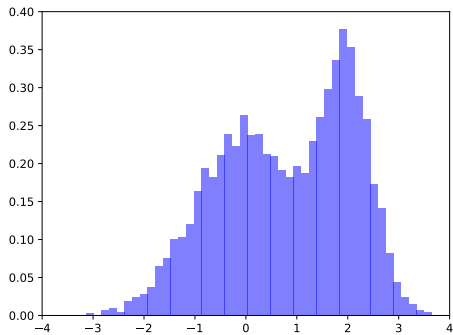
Intuition



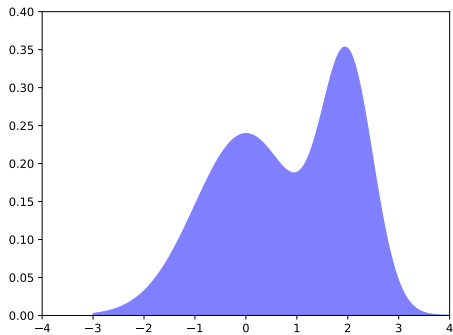
Intuition



Intuition

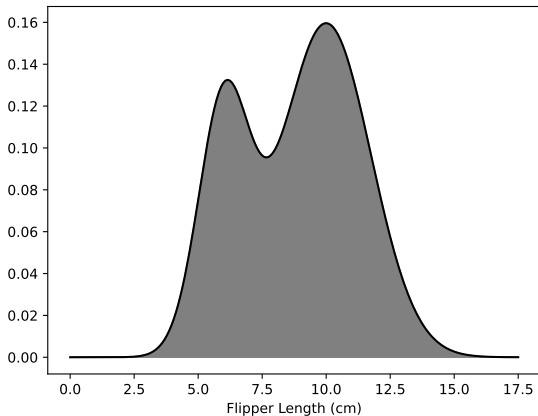


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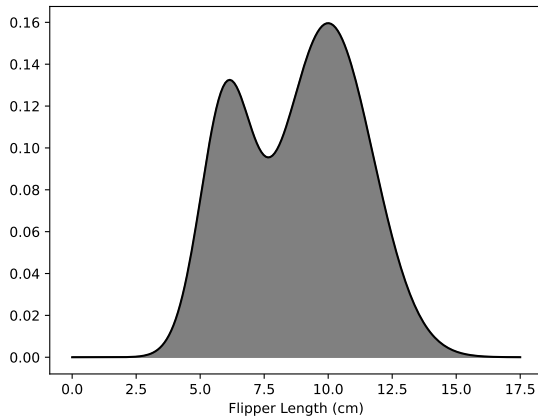
Marginal in X

- The distribution of flipper lengths is described by a density function, $p_X(x)$.



Exercise

What is the probability that Nature generates a penguin with flipper length equal to 10 cm?

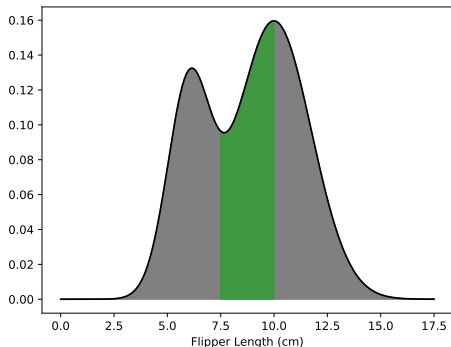


Solution

- ▶ **Zero!**
- ▶ $p_X(x)$ is **not** the probability that $X = x$.
- ▶ Instead, $\mathbb{P}(X = x) = \mathbb{P}(x < X < x) = \int_x^x p_X(x) dx = 0$
- ▶ The **probability** of a continuous random variable being *exactly* a certain value is zero.

Example

- What is the probability that Nature generates a penguin whose flipper length is between 7.5 and 10 cm?

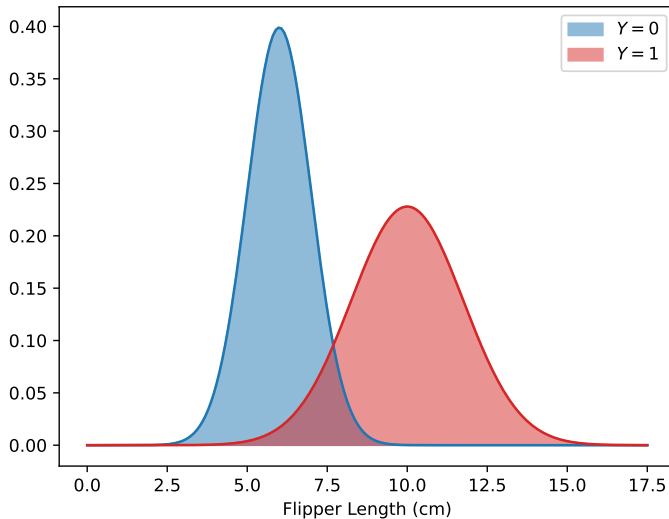


$$\mathbb{P}(7.5 < X < 10) = \int_{7.5}^{10} p_X(x) dx$$

Conditional on Y

- ▶ What is the probability of a certain flipper length, given that the species is y ?
- ▶ Also a continuous distribution, described by **conditional density** $p(x \mid Y = y)$.
- ▶ Two conditional density functions: one for $Y = 0$ and one for $Y = 1$.
 - ▶ Each integrates to one.

Conditional on Y



Conditional on X

- ▶ What is the probability that the species is y given a flipper length of x ?
- ▶ The conditional distribution of Y given X .

Exercise

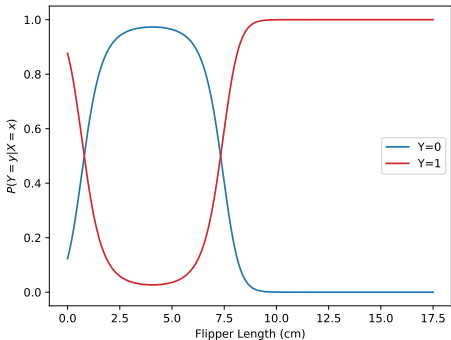
Is this distribution **continuous** or **discrete**?

Conditional on X

- ▶ Answer: **discrete**, because Y is discrete.
- ▶ One distribution $P(Y = y \mid X = x)$ for each possible value of X (infinitely many).

Conditional on X

- Although for any fixed x , $\mathbb{P}(Y = y \mid X = x)$ is discrete, we can plot the functions $f_0(x) = \mathbb{P}(Y = 0 \mid X = x)$ and $f_1(x) = \mathbb{P}(Y = 1 \mid X = x)$



Bayes' Rule

- Bayes' Rule applies to densities, too:

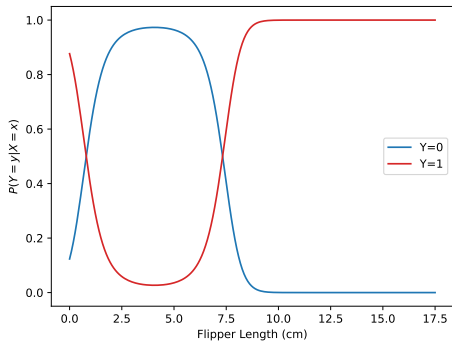
$$\mathbb{P}(Y = y \mid X = x) = \frac{p(x \mid Y = y)\mathbb{P}(Y = y)}{p_X(x)}$$

Bayes Decision Theory

- ▶ The Bayes classification rule applies to continuous distributions, too.
- ▶ **“Original” form:**
 - ▶ Predict class 1 if $\mathbb{P}(Y = 1 \mid X = x) > \mathbb{P}(Y = 0 \mid X = x)$;
 - ▶ Otherwise, predict class 0.
- ▶ **“Alternative” form:**
 - ▶ Using Bayes’ rule,
$$\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(X = x \mid Y = y)\mathbb{P}(Y = y) / \mathbb{P}(X = x) \dots$$
 - ▶ Predict class 1 if
$$\mathbb{P}(X = x \mid Y = 1)\mathbb{P}(Y = 1) > \mathbb{P}(X = x \mid Y = 0)\mathbb{P}(Y = 0)$$
 - ▶ Otherwise, predict class 0.

Exercise

Penguins with flippers of length 0, 3, and 12 are observed. What are their predicted species according to the Bayes' classification rule?



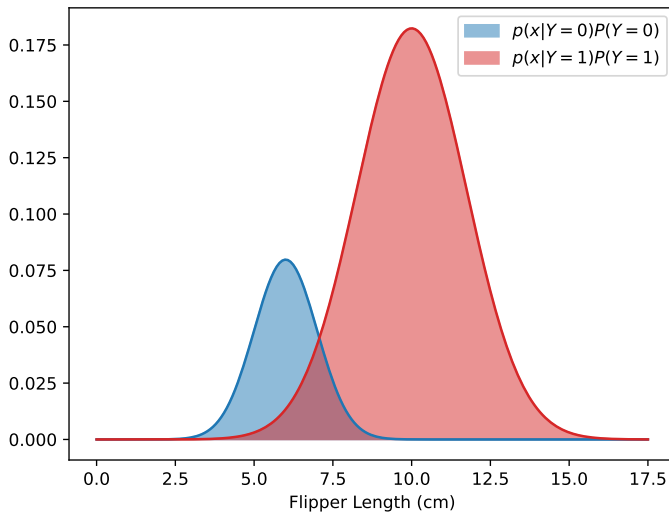
Joint

- ▶ The **joint** distribution in this case is neither totally continuous nor totally discrete.
- ▶ From Bayes' rule:

$$p(x, 0) = p(x \mid Y = 0)\mathbb{P}(Y = 0)$$

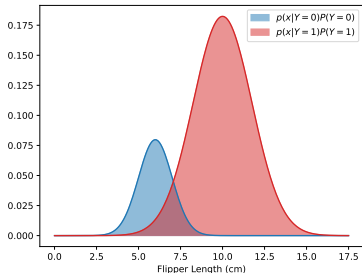
$$p(x, 1) = p(x \mid Y = 1)\mathbb{P}(Y = 1)$$

Joint Distribution



Exercise

Where does the Bayes decision rule make a prediction for class 1?



- Predict class 1 if $p(x | Y = 1)P(Y = 1) > p(X = x | Y = 0)P(Y = 0)$
- Otherwise, predict class 0.

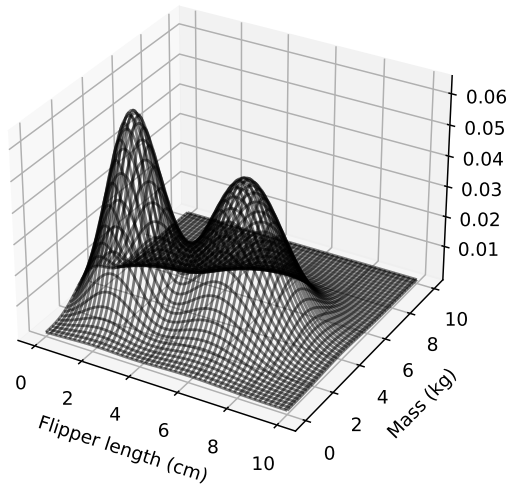
Multivariate Distributions

- ▶ In binary classification, $y \in \{0, 1\}$.
- ▶ But we usually deal with feature vectors, $\vec{x}$.
- ▶ The previous applies with straightforward changes.

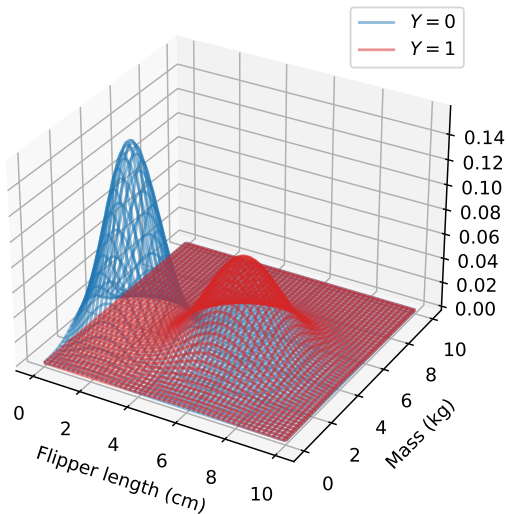
Example: Penguins

- ▶ Again consider penguins of two species, but now consider both flipper length and body mass.
- ▶ Each penguin's measurements are a **random vector**: $\vec{X}$.
- ▶ Densities are now functions of a vector.
 - ▶ E.g., marginal: $p_X(\vec{x}) : \mathbb{R}^2 \rightarrow \mathbb{R}^+$

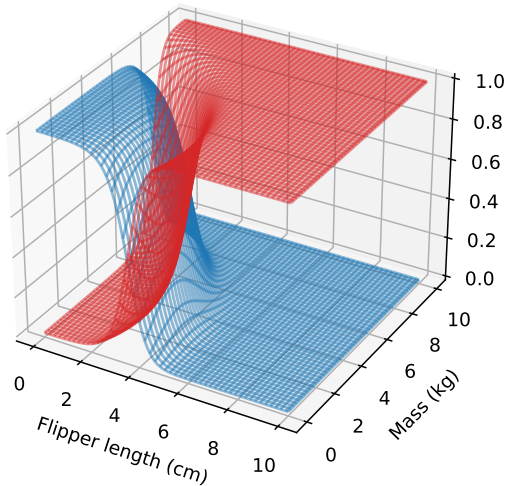
Marginal in $\vec{X}$



Conditional on Y



Conditional on X



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Lecture 11 | Part 4

Bayes Error

Bayes Error

- ▶ The **Bayes classification rule** is a natural approach to making predictions.
- ▶ It is also the **best you can do**, in a sense.

Errors

- ▶ Let's say you have a rule for predicting Y given $\vec{X}$.
- ▶ In binary classification, there are two types of errors:
 - ▶ Predicted 0, but the right answer is 1 (Case 1).
 - ▶ Predicted 1, but the right answer is 0 (Case 2).
- ▶ The probability of making a classification error is:

$$\mathbb{P}(\text{error}) = \mathbb{P}(\text{Case 1}) + \mathbb{P}(\text{Case 2})$$

Probability of Error

- ▶ Case 1: you predict 0, but the right answer is 1.
- ▶ The probability of this error is:

$$\begin{aligned}\mathbb{P}(\text{Case 1}) &= \mathbb{P}(Y \text{ is actually } 1, \text{ predict } 0) \\ &= \mathbb{P}(Y \text{ is actually } 1) \times \mathbb{P}(\text{predict } 0 \mid Y \text{ is actually } 1)\end{aligned}$$

Probability of Error

- ▶ Case 2: you predict 1, but the right answer is 0.
- ▶ The probability of this error is:

$$\begin{aligned}\mathbb{P}(\text{Case 2}) &= \mathbb{P}(Y \text{ is actually } 0, \text{ predict } 1) \\ &= \mathbb{P}(Y \text{ is actually } 0) \times \mathbb{P}(\text{predict } 1 \mid Y \text{ is actually } 0)\end{aligned}$$

Probability of Error

- The total probability of a classification error is:

$$\begin{aligned} P(\text{error}) &= \mathbb{P}(\text{Case 1}) + \mathbb{P}(\text{Case 2}) \\ &= \mathbb{P}(Y \text{ is actually } 1) \times \mathbb{P}(\text{predict } 0 \mid Y \text{ is actually } 1) \\ &\quad + \mathbb{P}(Y \text{ is actually } 0) \times \mathbb{P}(\text{predict } 1 \mid Y \text{ is actually } 0) \end{aligned}$$

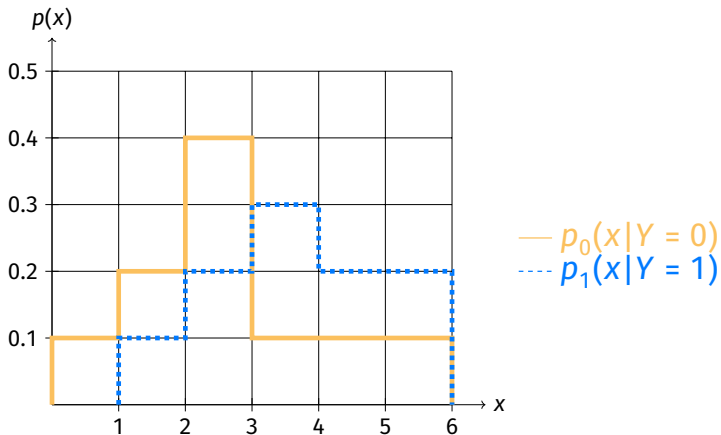
Fact

- ▶ The Bayes classifier has the smallest possible error probability² of any classifier.
- ▶ The error probability of the Bayes classifier is called the **Bayes error**.
- ▶ In most cases, the minimum possible error probability is >0 .

²Note that this is a statement about a probability, and not about any finite data set!

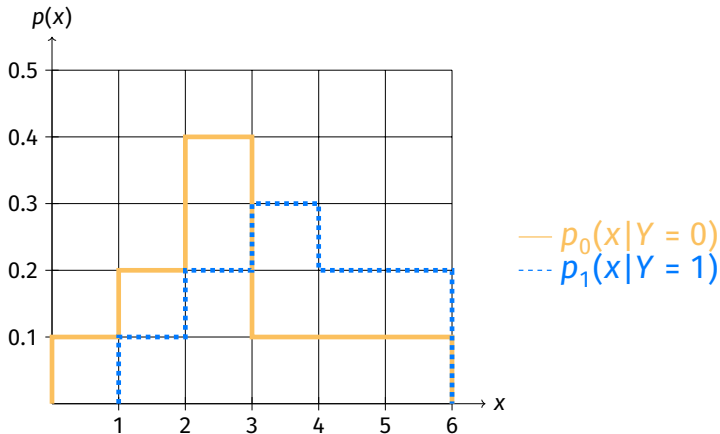
Exercise

What is the Bayes error if $P(Y = 0) = P(Y = 1) = 0.5$?

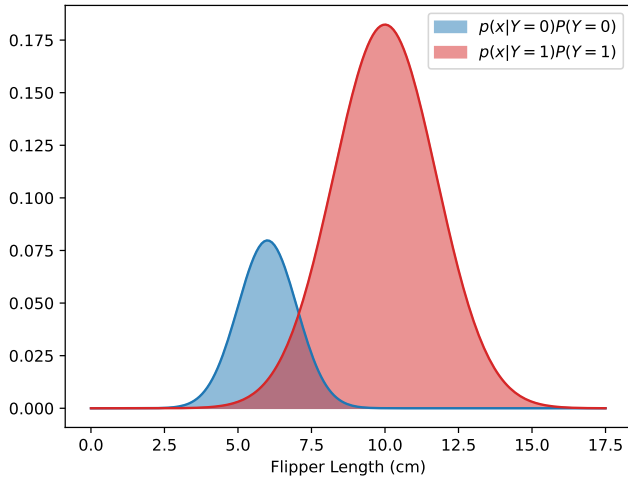


Exercise

What is the Bayes error if $P(Y = 0) = 0.3$ and $P(Y = 1) = 0.7$?



Intuition



Problem

- ▶ The Bayes classifier is optimal.
- ▶ But it requires knowing the joint distribution; we **almost never know this**.
- ▶ Next time: **estimating** probability distributions from data.